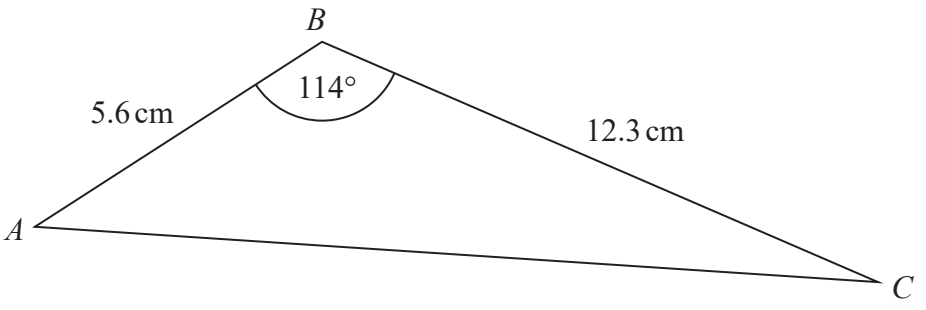
15 *ABC* is a triangle.



(a) Calculate the area of triangle *ABC*.
Give your answer correct to 3 significant figures.

Area = ½ × 5.6 × 12.3 × sin 114°

1 mark
Use included angle.

= 31.46250556... = 31.5 cm² (3sf)

1 mark
31.5 cm².

..... cm²

(2)

(b) Calculate the length of *AC*.
Give your answer correct to 3 significant figures.

$AC^2 = 5.6^2 + 12.3^2 - 2 \times 5.6 \times 12.3 \times \cos 114°$

= 182.65 – (–56.03203995...)

1 mark
Cosine rule.

= 238.68203995...

AC = √238.68203995... = 15.44933784... cm

1 mark
Subtract a negative.

= 15.4 cm (3 significant figures)

1 mark
15.4 cm.

..... cm

(3)

(Total for Question 15 is 5 marks)

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